

TULSA INTL, OK, USA

WMO: 723560

Lat: 36.199N

Lon: 95.887W

Elev: 650

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	18.2	1.5	6.1	17.1	5.9	7.6	22.0	26.6	50.1	24.8	47.9	9.4	350	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	100.1	75.5	97.2	76.2	94.6	76.0	79.6	92.6	78.5	91.2	77.6	90.1	10.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	139.7	86.2	75.0	134.3	85.2	74.0	129.8	84.2	43.6	92.7	42.4	91.5	41.4	90.0	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.8	19.5	DB	5.9	103.6	5.6	3.8	1.8	106.4	-1.5	108.6	-4.7	110.7	-8.7	113.5	(4)
(5)			WB	4.6	81.2	5.3	1.5	0.8	82.2	-2.3	83.1	-5.3	83.9	-9.1	85.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	38.6	42.8	51.9	61.0	69.8	78.6	83.3	82.4	74.2	62.6	51.1	41.0	(6)
(7)		DBStd	17.90	11.02	11.43	10.86	8.67	7.45	5.58	5.45	5.62	7.53	8.80	9.57	10.05	(7)
(8)		HDD50	1172	378	253	107	14	1	0	0	0	0	10	101	308	(8)
(9)		HDD65	3411	817	624	421	175	40	1	0	0	14	151	425	743	(9)
(10)		CDD50	5388	26	51	166	344	616	859	1031	1005	726	401	133	30	(10)
(11)		CDD65	2152	0	2	15	54	190	409	566	540	289	78	8	1	(11)
(12)		CDH74	24611	2	20	131	475	1702	4588	7313	6693	2973	656	57	1	(12)
(13)		CDH80	11758	0	4	24	99	535	2087	3957	3576	1293	178	5	0	(13)
(14)	Wind	WSAvg	9.3	9.4	9.7	10.6	11.2	9.9	9.5	8.3	7.7	7.7	8.8	9.5	9.1	(14)
(15)	Precipitation	PrecAvg	40.5	1.5	1.8	3.2	4.0	5.6	4.6	3.5	3.2	4.2	3.6	2.8	2.4	(15)
(16)		PrecMax	69.9	3.7	5.7	11.9	10.4	14.8	9.8	11.4	8.8	18.8	9.3	7.3	8.7	(16)
(17)		PrecMin	26.9	0.0	0.2	0.1	0.4	1.2	0.6	0.1	0.0	0.1	0.2	0.0	0.2	(17)
(18)		PrecStd	9.4	1.1	1.3	2.1	2.1	2.9	2.4	2.8	1.9	3.3	2.4	2.1	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	76.8	83.0	87.1	91.8	98.5	103.9	105.1	100.2	89.6	79.7	71.9	(19)
(20)			MCWB	55.8	60.3	63.6	66.3	72.8	75.9	75.0	74.7	73.7	71.9	64.1	61.9	(20)
(21)		2%	DB	66.2	71.0	77.4	82.8	88.4	95.1	100.1	100.3	94.2	84.9	74.8	66.2	(21)
(22)			MCWB	54.6	57.4	60.8	65.9	72.1	76.2	75.9	75.4	74.5	69.2	61.4	56.5	(22)
(23)		5%	DB	61.4	66.2	73.4	79.4	85.8	92.4	97.2	97.5	91.0	81.0	70.8	61.5	(23)
(24)			MCWB	51.7	55.0	59.8	64.3	71.4	75.4	76.7	75.9	74.0	66.5	60.2	53.3	(24)
(25)		10%	DB	56.1	61.5	69.3	76.1	82.8	90.0	94.6	94.5	87.6	77.3	66.9	56.8	(25)
(26)			MCWB	47.1	51.7	58.1	63.1	70.2	74.9	76.7	75.6	72.3	65.5	57.4	49.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.8	64.7	67.2	71.9	76.5	80.4	81.2	80.8	78.3	74.7	68.3	64.4	(27)
(28)			MCD B	67.7	71.6	78.0	80.3	86.4	92.2	93.8	93.5	90.6	84.9	75.1	70.0	(28)
(29)		2%	WB	57.4	60.9	64.6	69.5	74.8	78.6	79.9	79.1	76.8	72.6	64.6	59.2	(29)
(30)			MCD B	62.4	67.2	73.8	77.6	84.4	90.3	93.2	92.2	88.7	80.6	70.9	64.3	(30)
(31)		5%	WB	52.5	56.9	62.3	67.8	73.3	77.4	78.8	78.0	75.5	70.6	62.0	54.8	(31)
(32)			MCD B	59.9	64.5	70.8	75.9	83.0	88.9	91.8	91.2	86.7	77.2	68.4	59.1	(32)
(33)		10%	WB	47.5	51.8	59.5	65.5	71.7	76.2	77.8	77.0	74.2	67.8	59.0	50.2	(33)
(34)			MCD B	55.3	60.3	67.9	73.6	80.9	86.9	90.4	89.8	84.3	74.9	65.6	55.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.6	20.5	20.9	20.9	18.8	18.3	19.0	19.9	20.5	21.0	20.5	18.4	(35)
(36)			MCDBR	27.0	27.5	26.0	24.4	20.7	20.0	21.3	22.9	23.0	23.8	25.0	24.8	(36)
(37)			MCWBR	17.1	17.2	14.6	12.8	9.4	7.3	6.3	6.6	8.1	11.3	15.0	16.4	(37)
(38)		5% WB	MCDBR	23.4	24.5	21.0	20.2	18.3	18.4	18.9	19.3	19.7	18.4	21.5	20.6	(38)
(39)			MCWBR	16.7	17.6	13.6	12.8	9.5	8.2	7.3	7.0	7.6	10.5	15.6	16.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.309	0.354	0.387	0.426	0.433	0.443	0.444	0.396	0.351	0.317	0.298	(40)	
(41)		taud	2.488	2.484	2.391	2.316	2.268	2.279	2.294	2.255	2.363	2.470	2.510	2.515	(41)	
(42)		Ebn at Noon	285	296	290	284	273	270	266	264	272	277	277	279	(42)	
(43)		Edh at Noon	26	30	36	40	43	43	42	43	36	30	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	812	999	1315	1643	1815	2079	2091	1891	1581	1181	865	688	(44)	
(45)		RadStd	80	150	78	104	171	198	123	126	130	147	105	82	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	(46)
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page